

Belise Swartwood

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EDUCATION

Massachusetts Institute of Technology, Cambridge, MA May 2028
Bachelor of Science in Mathematics and Finance
Relevant Coursework: Probability & Statistics, Real Analysis, Discrete mathematics, Laboratory in Investments, Corporate Financial Accounting, Taxes & Business Strategy, Laboratory in Corporate Finance

SKILLS

Technologies: Python, R, MATLAB, Excel, Bloomberg Terminal
Technical: Financial statment analysis, FX & exchange rate dynamics, DCF Modeling, monetary policy analysis

EXPERIENCE

Central Bank of Armenia (CBA) June 2026 – Aug 2026
MISTI Eurasia Summer Intern - Dilijan, Armenia

- Analyzed the relationship between exchange rate volatility, monetary policy, and financial stability in an emerging market context
- Modeled optimal foreign currency deposit allocation using the Minimum Variance Portfolio framework applied to long-term AMD/USD exchange rate and price data, estimating the equilibrium dollarization level that minimizes macroeconomic risk
- Authored a central bank research paper on financial dollarization dynamics, translating quantitative findings into policy recommendations on FX reserve management and monetary transmission mechanisms

Massachusetts General Hospital Feb 2025 – Dec 2025
ML Neuroscience Researcher - Boston, Massachusetts

- Developed LASSO regression model ($R^2 = 0.71$) identifying the top 8 predictors of clinically significant depression treatment response across 2 fMRI datasets
- Validated findings with a classification model (AUC = 0.84); results presented at the Gerontological Society of America Conference 2025
- First author of manuscript submitted to [Journal Name]; research covers brain connectivity network patterns in treatment-resistant depression

Growing Islands: Rebuilding Beaches and Protecting Coastlines Oct 2024 – Feb 2025
MIT Undergraduate Researcher - Cambridge, Massachusetts

- Designed and tested physical prototypes to minimize sediment displacement by 20% and optimize wave disruption by 15% through controlled lab experiments and numerical simulations
- Trained ML model on coastal imagery to classify and predict sediment transport patterns, informing prototype design iterations

PROJECTS

Macro Investment Memo: USD/TRY Trade in a Higher-for-Longer Rate Environment June 2026

- Authored an independent macro research memo recommending a long USD/TRY position, building yield curve, real yield, and inflation-differential models in Python from FRED, BLS, and CME FedWatch data
- Tested the conventional thesis that rising US real yields drive EM currency weakness, identified the Turkey-US inflation differential as the dominant driver (0.75 correlation since 2018), constructed a trade with explicit entry, target, stop-loss, and risk scenarios

Merger Arbitrage Investment Strategy Research Nov – Dec 2025

- Designed and backtested the “ArbiTIM” statistical arbitrage strategy across 285 all-cash M&A deals (\$500M+) from 2015–2024; constructed a market-neutral portfolio achieving low beta and reduced volatility relative to the S&P 500
- Quantified deal-failure probabilities and modeled regulatory execution risk using R and Bloomberg Terminal; optimized position sizing based on risk-adjusted return profiles and tail-risk scenarios

ACTIVITIES

- MIT Varsity Swimming & Diving
 - NCAA DIII Finalist, 2025 Scholar All-American
- MIT Chamber Music Society; MIT Emerson/Harris Scholar, Violinist
- President, MIT Hawaii Club
- Interests: surfing, running, drawing